



PRODUCT OUTLINE

3G AFE - Analog Baseband

PART NUMBER:

CI7640tm

FEATURES:

- 90nm standard CMOS
- Dual Voltage supply:
 - I/O 2.85V
 - Analog and Digital 1.2+/-10%
- I/Q Rx path
 - Two fully parallel paths for diversity
 - Supports CDMA/WCDMA/EGPRS
 - Programmable gain
 - 12-bit $\Sigma\Delta$ ADCs with 60dB DR for CDMA/WCDMA and 80dB for EGPRS
- I/Q Tx path
 - Supports CDMA/WCDMA/EGPRS
 - Highly oversampled 10-bit DAC
 - Current and voltage output modes
- Auxiliary general purpose ADC
 - 10-bit 400kS/s
 - 6 multiplexed inputs
- Auxiliary Radio Control DACs
 - 12-bit 40us for AFC
 - 10-bit 10us for AGC
 - 10-bit 10us for general purpose
 - 10-bit 0.5us for PA control
- Internal references including
 - Bandgap voltage
 - Reference current
- Typical Current consumption

Each I/Q Rx analog path:

 - EGPRS: 5 mA
 - CDMA: 8 mA
 - WCDMA: 12 mA

- I/Q Tx analog path: 9mA

Auxiliary DACs (total): 2.5mA
- Core Area < 4 mm²

APPLICATIONS:

- 3G Cellular Phones and PC-Cards
- Highly integrated Baseband Processors

OVERVIEW

CI7640tm is a fully integrated analog baseband processor that is capable of performing all the functions necessary for the DSP/Radio interface in EGPRS, CDMA and WCDMA applications.

Very compact area makes it suitable for embedded integration together with the digital baseband SOC.

The Radio Control section includes several auxiliary DACs for DSP/Radio interface in GSM/CDMA/WCDMA applications.

An on-chip 10-bit ADC with multiplexed inputs allows to monitor parameters such as RSSI, battery voltage, temperature, etc.

This device operates from 2.6V to 3.3V and is directly controlled by a digital baseband modem through a serial interface.

The analog interfaces are highly programmable to allow interfacing with a wide variety of RF chip-sets.

All necessary voltage references and biasing currents are generated internally with only the need of some extra pins for decoupling.

A power-down mode is available with less than 15uA current consumption.

BLOCK DIAGRAM

